

RADHEY KALRA

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Full-stack developer crafting ML/audio systems and self-hosted infrastructure.

EDUCATION

Bachelor of Technology in CSE

Model Institute of Engineering and Technology

Aug. 2023 – May 2027

Jammu and Kashmir, India

Schooling (Class I – XII)

Air Force School, Jammu

2011 – 2023

Jammu and Kashmir, India

TECHNICAL SKILLS

Languages: Python, C/C++, TypeScript/JavaScript, Go, Vala, SQL, Rust, Bash, PowerShell

Frameworks & Platforms: SvelteKit, React, Next.js, Astro, Tailwind CSS, React Native, Expo, Bun, Node

Developer Tools: Git, Docker, CMake, CI/CD, Cloudflare, Vercel, Kubernetes, Terraform

Databases & Cloud: Convex, PostgreSQL, MySQL, AWS, [GCP](#)

AI/ML Libraries: PyTorch, TensorFlow, Transformers, scikit-learn, NumPy, Pandas, Matplotlib, Seaborn

Domains: Deep Learning, Audio Processing, Linux Servers, Full-Stack Development, Networking

PROJECTS

HPOMen Linux (Victus Control)

Mar. 2026 – Present

Reverse-engineering HP OMEN control utility behavior on Linux | [GitHub](#) | [AUR](#)

- Developing a Linux-first OMEN/Victus control suite in **Vala** with a GTK4 monitor app, tray companion, D-Bus helper, and probe CLI.
- Reverse-engineered **HP WMI/sysfs** behavior to validate profile/fan controls (**auto/max**) and extend **hp_wmi** with capability-gated safety fallbacks.
- Proposed and upstreamed a patch to platform driver x86 in the Linux kernel to expose missing sysfs controls.

BoxBox

Nov. 2025 – Present

Self-hosted file management system for Linux servers | [Website](#) | [GitHub](#)

- Built a **lightweight** self-hosted file manager in **Svelte + Go** with multi-mount browsing, chunked transfers, and real-time updates.
- Added media preview/stream/download support and **Monaco Editor** integration for in-browser text file editing.

U-Net Audio Filter

Jul. 2025 – Present

Audio enhancement by removing background noise | [GitHub](#)

- Building an audio enhancement system with a **custom U-Net-inspired architecture** and end-to-end audio denoising pipeline.
- Implemented **Python/PyTorch + audio DSP** preprocessing, spectrogram modeling, and evaluation tooling, with early results showing minor gains over open-source baselines.

Fast Blob Storage (FBS)

Apr. 2026 – May 2026

S3-compatible blob storage and CDN solution primarily for self-hosting | [Live](#) | [GitHub](#) | [Docker](#)

- Built a single-binary **Go** storage backend with **SQLite** metadata, decoupled **SvelteKit** dashboard, and a published Docker image.
- Supported Both SigV4 and bearer token authentication, using linux **sendfile()** for efficient zero-copy file transfers.
- Achieved binary sizes of **15MB** and idel RSS of **16MB** and under load RSS of **21MB** in controled benchmarks, making it suitable for self-hosting on resource-constrained environments.

Additional Projects

- **QckPages** — SvelteKit/Convex app for publishing HTML pages, FBS-backed storage ([Live](#) | [GitHub](#))
- **rsclip** — Rust Wayland clipboard manager with a low-memory daemon, GTK4 UI, SQLite history, search, filters, OCR Support and Secrets management ([GitHub](#) | [AUR](#))
- **Homelab Infrastructure** — Infrastructure as Code (IaC) for self-hosted server storage, remote access, and service deployment ([GitHub](#))
- **Dots-Niri** — Replicates my full NiriWM rice on Arch-based systems with a single **curl** command and a guided one-shot TUI installer workflow ([Website](#) | [GitHub](#))
- **WallPy-Gui** — Wallpaper GUI app for Wayland with support for image, GIF, and video formats ([GitHub](#) | [AUR](#))
- **iwd-applet** — Wi-Fi tray applet for menu-driven wireless management using **iwd** on Linux ([GitHub](#))
- **HowTo** — Astro tutorials site for practical hardware and software walkthroughs ([Live](#) | [GitHub](#))

- **Pico32** — C++/PlatformIO serial TUI for ESP32 Wi-Fi and Bluetooth pentesting workflows ([Website](#) | [GitHub](#))
- **TL;DR** — Electron app that summarizes WhatsApp chats with OpenRouter compatible AI models ([GitHub](#))
- **Community Leaderboard** — Next.js dashboard for event tracking, rankings, and community participation insights ([Live](#) | [GitHub](#))
- **Hotel Booking** — Svelte/Express/MySQL booking platform with authentication and admin workflows ([GitHub](#))
- **Tenant Manager** — React Native app for managing tenants and tracking electricity usage ([GitHub](#))
- **Terminal Doom** — Doom-inspired terminal game built with C++, `ncurses`, and `CMake` ([GitHub](#))

LEADERSHIP, CO-CURRICULAR ACTIVITIES

Cloud Lead

Aug. 2025 – Present

GDGoC (Google Developer Groups on Campus), MIET Jammu

Jammu and Kashmir, India

- Led cloud learning initiatives through hands-on **Google Cloud** workshops and guided sessions for **100+ students**, and built a tutorial platform ([Live/GitHub](#)) with **20+** lab walkthroughs, notes, and video resources to support **Google Cloud Study Jam** completion.

Cloud and DevOps Lead

Jan. 2026 – Present

AWS Cloud Club MIET Jammu

Jammu and Kashmir, India

- Lead cloud and DevOps initiatives through hands-on workshops, technical sessions, and collaborative projects, driving adoption of **AWS**, containerization, and infrastructure best practices focused on deployment automation and scalability.

IDE Bootcamp Edition 3 Team Leader – 2nd Place

Apr. 2026

Innovation, Design, and Entrepreneurship (IDE) Bootcamp

GNDEC, Ludhiana, Punjab

- Led a team to secure **2nd Place** at the AICTE/MoE Innovation Cell bootcamp by applying rapid prototyping and business modeling to develop an AI-powered B2B SaaS product for automated presentation generation from user-uploaded templates and documents.

IDE Bootcamp Edition 2 Team Leader

Feb. 2025

Innovation, Design, and Entrepreneurship (IDE) Bootcamp

IIT Jammu

- Participated in a 5-day immersive bootcamp at IIT Jammu with Wadhvani Foundation, AICTE, and MoE's Innovation Cell, gaining business modeling, product design, and entrepreneurship experience while leading a team to develop a "Sports Match Finder" app using design thinking and the Lean Canvas model.

Hackathon Team Leader

Internal Hackathon & Smart India Hackathon (SIH 2024 & 2025)

MIET, Jammu

- Led multidisciplinary teams in MIET's internal hackathon and Smart India Hackathon selection rounds in **2024** and **2025**, reaching the **top 9 teams in the college** both years through collaborative problem-solving, rapid prototyping, and real-world solution design.

CERTIFICATIONS & PROGRAMS

Google Cybersecurity - Coursera | [Certificate](#)

Cisco Cybersecurity Operations Fundamentals - Coursera | [Certificate](#)

EC-Council Cybersecurity Attack and Defense Fundamentals - Coursera | [Certificate](#)

Palo Alto Networks Cybersecurity - Coursera | [Certificate](#)

Nptel Ethical Hacking - IIT Kharagpur | [Certificate](#)

Nptel Deep Learning - IIT Ropar | [Certificate](#)

Nptel Artificial Intelligence for Management | [Certificate](#)

IBM SkillsBuild: AI Agent Architect | [Certificates](#)

IDE Bootcamp (AICTE / MoE) | [Certificate](#)